RESEARCH ANALYSIS REPORT

Bitcoin and Selected Cryptocurrency Correlations

Hourly opening-price levels and high-to-open returns

This report summarizes two exploratory Pearson correlation analyses based on canonical Binance Spot hourly klines quoted in USDT. The eligible universe contains pairs whose first observed hourly candle occurred in or before calendar year 2021 UTC.

ELIGIBLE PAIRS	PRICE-LEVEL RANGE	RETURN RANGE
23	**-0.195 to 0.942**	**0.022 to 0.697**
22 assets compared with BTC	Correlation with BTC	Correlation with BTC

Executive findings

- Opening-price levels produced five strong positive correlations using the descriptive 0.70 threshold. BNB ranked highest at 0.9418; GALA ranked lowest at -0.1947.
- High-to-open returns produced no coefficient at or above 0.70. LTC ranked highest at 0.6972, followed closely by ETH at 0.6934; UNI ranked lowest at 0.0222.
- The rankings changed materially between the analyses. High price-level correlation does not imply equally high correlation in the defined intrahour return.
- Neither analysis establishes causality or predictive power. Pair-specific overlap periods and preserved gaps must accompany interpretation.

Analysis design

Element	Opening-price analysis	High-to-open return analysis
Variable	Hourly open price level in USDT	(high - open) / open for each hourly candle
Method	Pearson correlation	Pearson correlation
Alignment	Exact UTC opening timestamps; pairwise complete	Exact UTC opening timestamps; pairwise complete
Missing data	No interpolation or filling	No interpolation or filling
Period	Pair-specific starts through 2026-09-01 23:00 UTC	Pair-specific starts through 2026-09-01 23:00 UTC

1. Opening-price level correlation

This analysis measures synchronous linear association between hourly opening-price levels. The full matrix below uses pairwise-complete UTC observations, so each cell can represent a different overlap period.

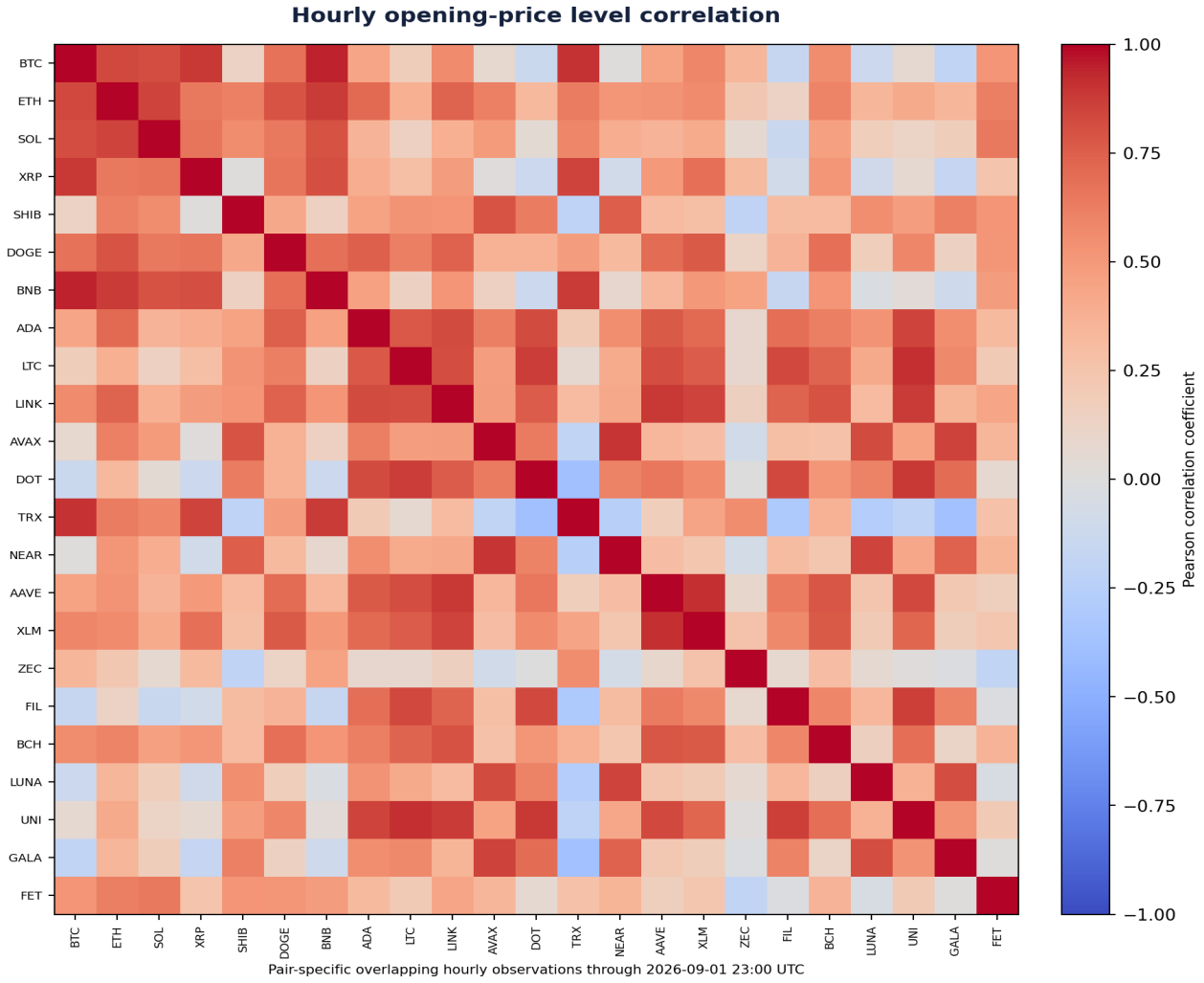


Figure 1. Pearson correlation matrix for hourly opening-price levels. The coefficient is dimensionless.

Correlation bands against BTCUSDT

Descriptive band	Count	Share
Strong positive	5	22.7%
Moderate positive	7	31.8%
Weak positive	6	27.3%
Negative	4	18.2%

Interpretation: five assets - BNB, TRX, XRP, ETH, and SOL - exceeded the 0.70 descriptive threshold. However, correlations between trending, potentially non-stationary price levels may be inflated by common market trends.

Opening-price relationship with Bitcoin

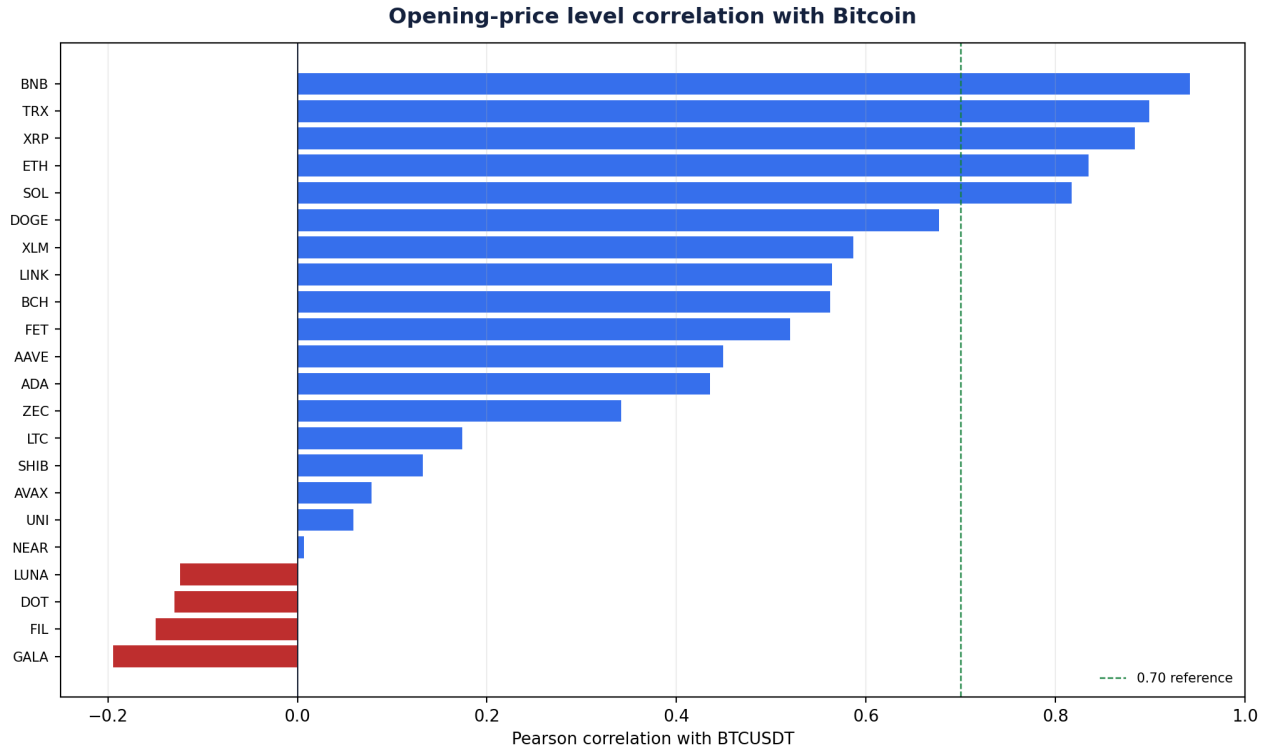


Figure 2. Eligible assets ranked by opening-price level correlation with BTCUSDT.

Highest and lowest coefficients

Asset	Correlation	Paired hours	Band
BNBUSDT	0.9418	77,203	Strong positive
TRXUSDT	0.8990	72,021	Strong positive
XRPUSDT	0.8839	72,936	Strong positive
ETHUSDT	0.8348	79,140	Strong positive
SOLUSDT	0.8170	53,086	Strong positive
NEARUSDT	0.0066	51,551	Weak positive
LUNAUSDT	-0.1240	52,405	Negative
DOTUSDT	-0.1297	52,901	Negative
FILUSDT	-0.1493	51,515	Negative
GALAUSDT	-0.1947	43,551	Negative

Across all 22 comparisons, the median price-level correlation with Bitcoin was 0.4425. Paired sample sizes ranged from 43,551 to 79,140 hourly observations.

2. High-to-open return correlation

The second analysis transforms every candle using (high - open) / open. This non-negative measure captures the maximum observed appreciation within the hour relative to the opening price.

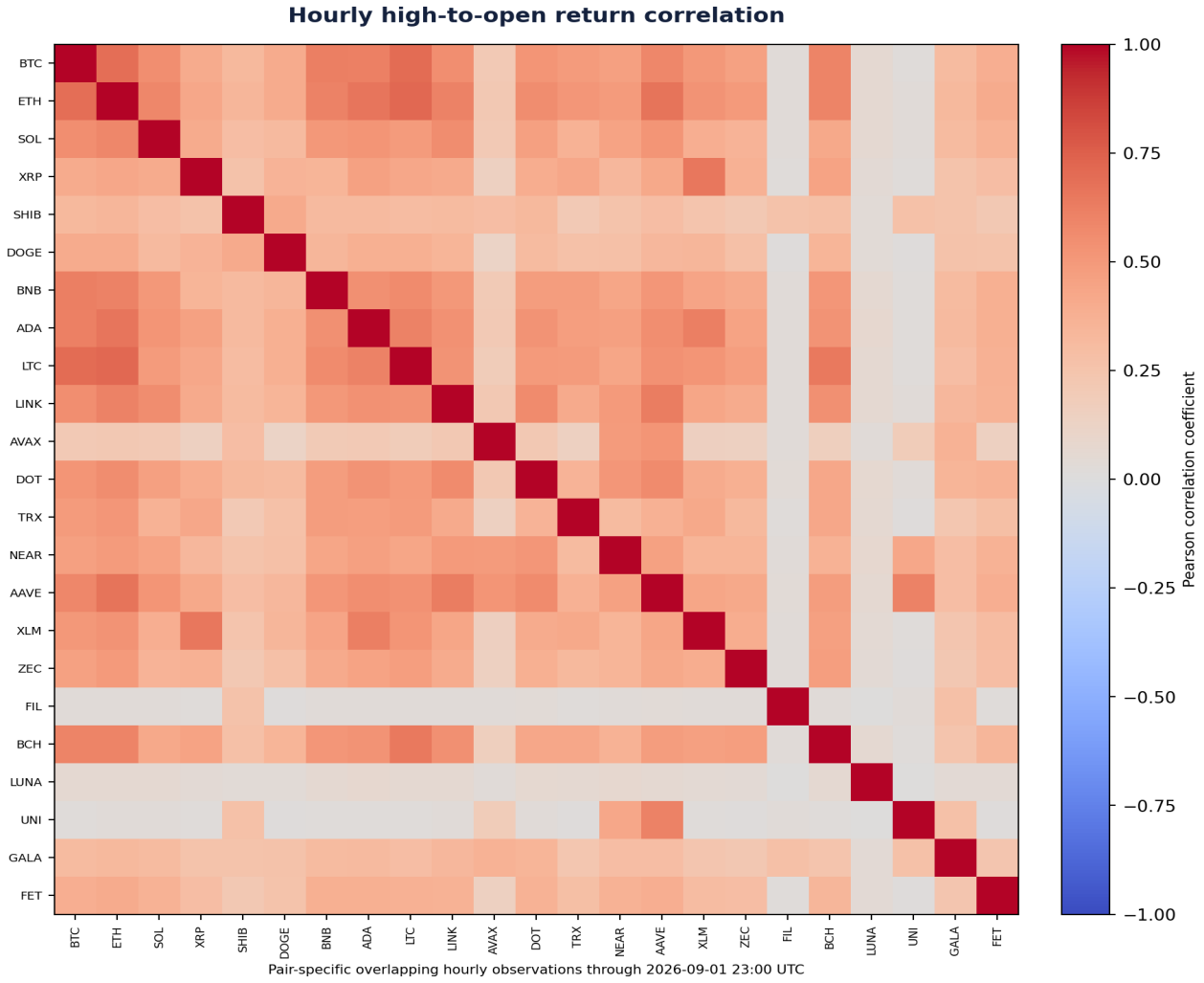


Figure 3. Pearson correlation matrix for hourly high-to-open returns. Returns are expressed as decimal ratios.

Correlation bands against BTCUSDT

Descriptive band	Count	Share
Strong positive	0	0.0%
Moderate positive	14	63.6%
Weak positive	8	36.4%
Negative	0	0.0%

Interpretation: all 22 coefficients were positive, but none reached 0.70. The pattern is predominantly moderate or weak rather than uniformly strong.

High-to-open return relationship with Bitcoin

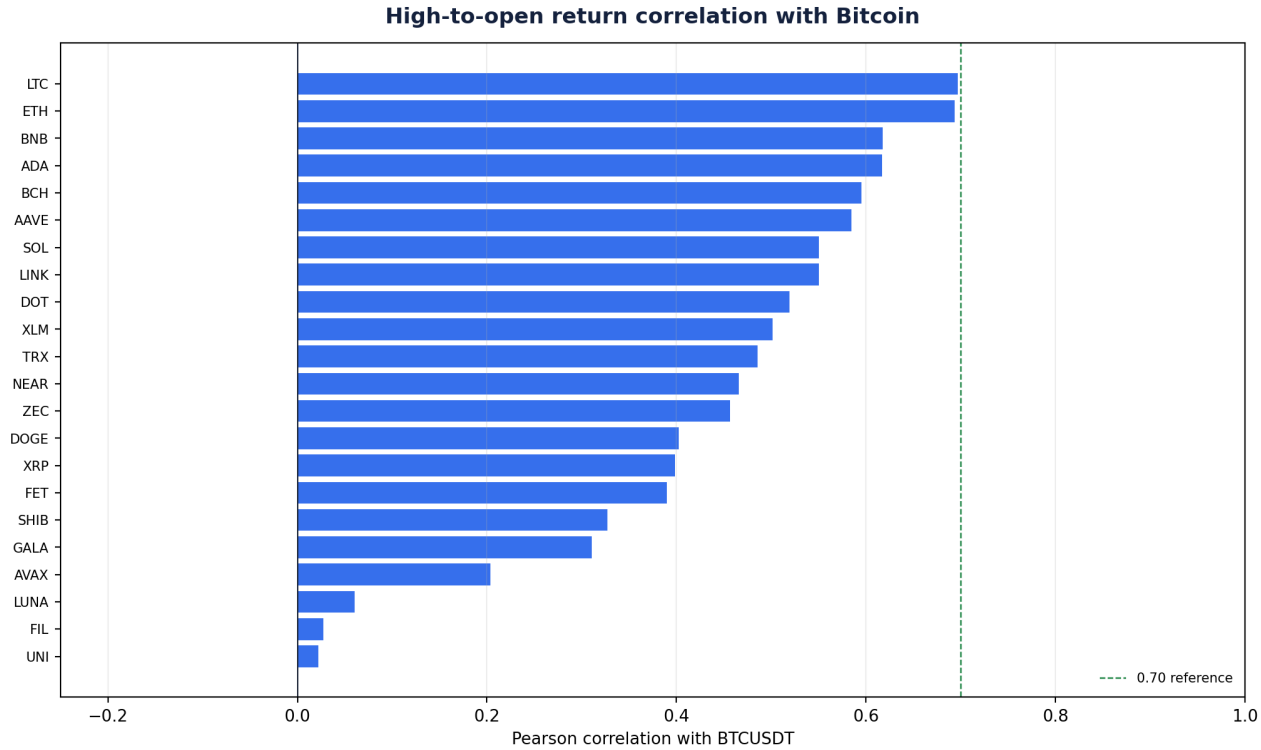


Figure 4. Eligible assets ranked by high-to-open return correlation with BTCUSDT.

Highest and lowest coefficients

Asset	Correlation	Paired hours	Band
LTCUSDT	0.6972	76,315	Moderate positive
ETHUSDT	0.6934	79,140	Moderate positive
BNBUSDT	0.6180	77,203	Moderate positive
ADAUSDT	0.6167	73,348	Moderate positive
BCHUSDT	0.5951	59,238	Moderate positive
GALAUSDT	0.3103	43,551	Weak positive
AVAXUSDT	0.2036	52,078	Weak positive
LUNAUSDT	0.0605	52,405	Weak positive
FILUSDT	0.0276	51,515	Weak positive
UNIUSDT	0.0222	52,201	Weak positive

The median high-to-open return correlation with Bitcoin was 0.4757. The range was 0.0222 to 0.6972; no negative coefficient was observed.

3. Comparison and interpretation

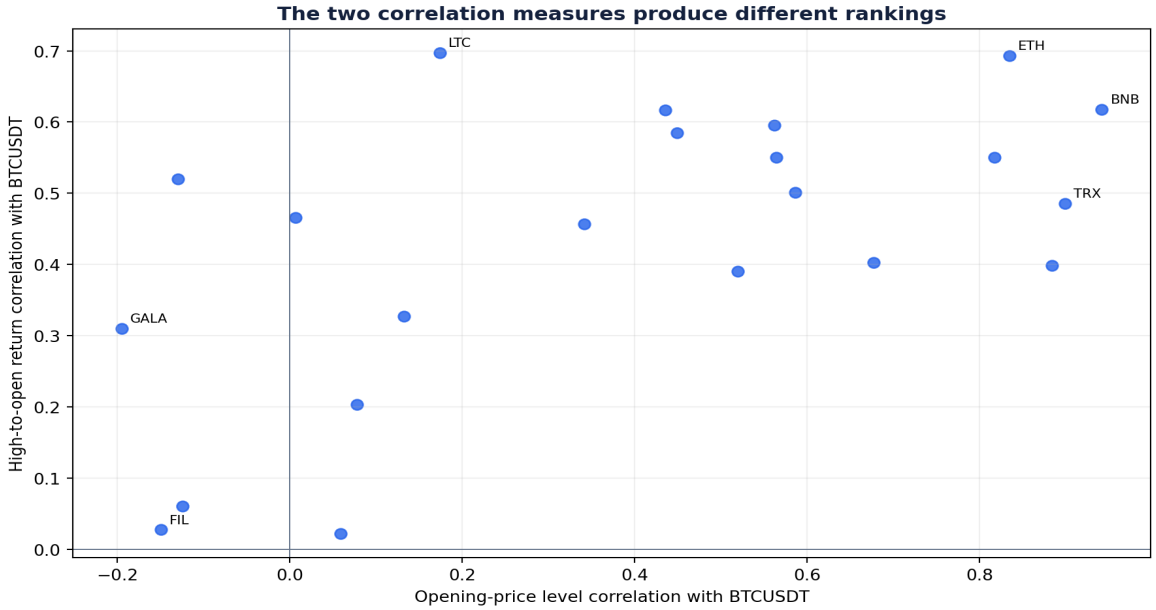


Figure 5. Each point represents one asset's two correlations with BTCUSDT.

Asset	Price level	High-to-open return	Change in rank
BNBUSDT	0.9418	0.6180	+2
TRXUSDT	0.8990	0.4857	+9
XRPUSDT	0.8839	0.3987	+12
ETHUSDT	0.8348	0.6934	-2
SOLUSDT	0.8170	0.5505	+2
DOGEUSDT	0.6768	0.4023	+8
XLMUSDT	0.5866	0.5012	+3
LINKUSDT	0.5640	0.5500	+0
BCHUSDT	0.5624	0.5951	-4
FETUSDT	0.5200	0.3901	+6
AAVEUSDT	0.4495	0.5845	-5
ADAUSDT	0.4355	0.6167	-8
ZECUSDT	0.3419	0.4568	+0
LTCUSDT	0.1740	0.6972	-13
SHIBUSDT	0.1326	0.3271	+2
AVAXUSDT	0.0779	0.2036	+3
UNIUSDT	0.0594	0.0222	+5
NEARUSDT	0.0066	0.4658	-6
LUNAUSDT	-0.1240	0.0605	+1
DOTUSDT	-0.1297	0.5196	-11
FILUSDT	-0.1493	0.0276	+0
GALAUSDT	-0.1947	0.3103	-4

Rank change is return-analysis rank minus price-level rank. A positive number means the asset ranked lower under the high-to-open return measure.

4. Methodological cautions

Correlation is not causation
Neither matrix demonstrates that Bitcoin directly determines another asset's price or return.

Price levels may be non-stationary
Common long-run trends can create high level correlations even when short-horizon co-movement is weaker.

The return definition is specialized
(high - open) / open measures maximum intrahour upside, not a conventional close-to-close return.

The candle high is ex-post
The hourly high is known only after the hour unfolds and cannot be used as information available at the candle opening.

Execution is not modeled
The observed high is not necessarily tradable without a rule for order placement, liquidity, slippage, fees, and fill probability.

Pearson is outlier-sensitive
Extreme candles can materially affect coefficients. Robustness analysis would require an approved extension.

Samples differ by asset
Pairwise-complete alignment preserves data but gives recently listed assets shorter and more recent comparison periods.

LUNA requires special care
Its documented May 2022 interruption and ecosystem migration create a material continuity risk.

Reproducibility

Item	Value
Data source	Canonical gzip-compressed Binance Spot hourly CSV files
Candidate universe	30 approved USDT-quoted pairs
Eligible universe	23 pairs with first observed candle in or before 2021 UTC
Overall timestamp envelope	2017-08-17 04:00 UTC through 2026-09-01 23:00 UTC
Missing-data policy	Pairwise complete; no interpolation or filling
Analysis notebooks	06_binance_hourly_open_price_correlation.ipynb; 07_binance_hourly_high_open_return_correlation.ipynb
Report generated	2026-09-05 19:46 UTC